

# **HELLO TAMIL – BILINGUAL WORD TRANSLATION MODULE.**

Project ID: 25-26J-281

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Department of  
Information Technology

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August 2025

## Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

| Name         | Student ID | Signature                                                                           |
|--------------|------------|-------------------------------------------------------------------------------------|
| Ahamed M A S | IT22229502 |  |

The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

The above candidate is carrying out research for the undergraduate dissertation under my supervision.

.....

Signature of the supervisor:

(Mr. Suresh Niroshan)

Date: .....

.....

Signature of the Co-supervisor:

(Ms. Fathima Fanoon)

Date: .....

## Abstract

The “Hello Tamil – Tamil Language Learning System for Non-Native Children in Sri Lanka” project aims to address the persistent challenge faced by Sinhala-speaking children in learning the Tamil language. This proposal focuses on one key module — the Bilingual Word Translation Module — which enables children to learn Tamil vocabulary through interactive, gamified activities and bilingual (Tamil–Sinhala/English) translation features.

The module integrates OCR to extract Tamil text from images, a Translation Engine (based on Google Translate API or a custom bilingual dictionary) for real-time Tamil-to-Sinhala/English translation, and TTS for pronunciation support. To enhance motivation, the system employs gamification techniques such as points, badges, daily challenges, and mini games (e.g., “2 Pictures 1 Meaning”) that make language learning playful and engaging.

This approach bridges the gap between technology and education, transforming language learning into a fun, self-directed experience. It leverages proven research in game-based learning, child-focused translation education, and multilingual vocabulary acquisition to create a system that improves both vocabulary retention and learner engagement. The proposed module will be developed as a mobile application (Android) and evaluated through technical testing and user studies with non-native children aged under 8 to assess usability, engagement, and learning gains.

By combining translation technology with gamified learning principles, the Bilingual Word Translation Module aspires to make Tamil language learning more accessible, motivating, and effective for non-native young learners in Sri Lanka.

**Keywords:** Tamil Learning, Gamification, Word Translation, OCR, TTS, Bilingual Education, Children Learning.

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## List of Abbreviations

| Abbreviation | Description                                      |
|--------------|--------------------------------------------------|
| AI           | Artificial Intelligence                          |
| API          | Application Programming Interface                |
| ASR          | Automatic Speech Recognition                     |
| CNN          | Convolutional Neural Network                     |
| DGBLL        | Digital Game-Based Language Learning             |
| GUI          | Graphical User Interface                         |
| L2           | Second Language                                  |
| ML           | Machine Learning                                 |
| NLP          | Natural Language Processing                      |
| NMT          | Neural Machine Translation                       |
| OCR          | Optical Character Recognition                    |
| SDK          | Software Development Kit                         |
| TTS          | Text-to-Speech                                   |
| UI           | User Interface                                   |
| UX           | User Experience                                  |
| WBS          | Work Breakdown Structure                         |
| XP           | Experience Points (used in gamification systems) |

# 1. Introduction

## 1.1 Background & Literature Survey

Sri Lanka is a multilingual country where both Tamil and Sinhala are official languages, yet most Sinhala-speaking children find it difficult to learn Tamil effectively at the primary school level. Safeek and Ponniah [3] pointed out that traditional teaching approaches rely heavily on memorization and textbook-based learning, offering little interactivity or contextual understanding. As a result, students often fail to connect Tamil vocabulary with their Sinhala or English equivalents, limiting bilingual communication skills.

The rise of digital and game-based learning presents new opportunities for improving L2 acquisition. Chowdhury and colleagues [8], along with Mei and co-authors [13] emphasized that DGBLL significantly improves motivation, vocabulary recall, and learning outcomes in children. Shortt and colleagues [9] further noted that gamified learning systems such as Duolingo successfully use points, levels, and streaks to sustain learner engagement.

Recent research also explores the use of AI and translation technologies for child-oriented learning. Husin and Ismail [7] introduced the “2 Pictures 1 Meaning (2P1M)” model, demonstrating that image-based translation games can improve comprehension and retention among students. Similarly, Dită et al. [1] proposed integrating OCR and augmented reality into language learning, allowing learners to translate printed text into digital formats for instant understanding.

NMT and OCR technologies have also matured enough to support low-resource languages like Tamil and Sinhala. Choudhary and colleagues [4], together with Bamotra and Uppala [5], demonstrated that these technologies can enhance bilingual translation effectiveness. Additionally, Liu and co-authors [2] developed the SingaKids project, which uses multimodal AI tutors combining text, images, and speech to support interactive vocabulary learning.

However, most existing Tamil learning applications are designed for Tamil speakers or English learners and do not provide Tamil–Sinhala translation. They also lack gamified features that motivate young learners. Krystalli and Arvanitis [10] highlighted these limitations, indicating a clear gap for a bilingual Tamil-to-Sinhala word translation tool specifically designed for non-native young learners.

## 1.2 Research Gap

Existing Tamil learning applications such as Tamil Kids Learning and Akuru Pahan focus on letter recognition and tracing, offering little or no bilingual support. Generic translation tools like Google Translate can convert Tamil words to Sinhala or English but are not designed for children, lacking interactive features, image association, or learning feedback Liu and colleagues [2], as well as Krystalli and Arvanitis [10], highlighted these limitations.

Previous research has successfully demonstrated gamified vocabulary systems and translation frameworks, yet none have been localized for Tamil–Sinhala bilingual learning. Bamotra and Uppala [5] integrated OCR with word translation, and Husin and Ismail [7] applied game-based translation for English learners. However, no existing system combines these technologies to create a child-friendly Tamil vocabulary learning solution.

Thus, the research gap lies in the absence of an interactive, gamified Tamil–Sinhala word translation system that uses text and image input, AI translation, and reward-based learning mechanisms to improve vocabulary acquisition among non-native children under 8 years old.

| Research Gap Feature                                | Husin & Ismail (2025) | Chowdhury et al. (2024)      | Bamotra & Uppala (2023) | Liu et al. (2025)       | Proposed Solution |
|-----------------------------------------------------|-----------------------|------------------------------|-------------------------|-------------------------|-------------------|
| Tamil–Sinhala bilingual translation                 | ✗                     | ✗                            | ✗                       | ✗                       | ✓                 |
| Gamified translation-based learning                 | ✓                     | ✓                            | ✗                       | ✗                       | ✓                 |
| Use of OCR for image-based word recognition         | ✗                     | ✗                            | ✓                       | ✗                       | ✓                 |
| TTS for pronunciation                               | ✗                     | ✗                            | ✗                       | ✓                       | ✓                 |
| Child-focused bilingual learning (under 8 years)    | ✓                     | ✓                            | ✗                       | ✓                       | ✓                 |
| Integration of NMT / AI translation                 | ✗                     | ✗                            | ✓                       | ✓                       | ✓                 |
| Evaluation with children (usability and engagement) | ✓                     | ✓                            | ✗                       | ✓                       | ✓                 |
| Gamified reward system (points, badges, word hunts) | ✓                     | ✓                            | ✗                       | ✗                       | ✓                 |
| Research Gap Feature                                | Mei et al. (2023)     | Krystalli & Arvanitis (2024) | Safeek & Ponniah (2024) | Choudhary et al. (2020) | Proposed Solution |
| Tamil–Sinhala bilingual translation                 | ✗                     | ✗                            | ✗                       | ✗                       | ✓                 |
| Gamified translation-based learning                 | ✓                     | ✓                            | ✗                       | ✗                       | ✓                 |
| Use of OCR for image-based word recognition         | ✗                     | ✗                            | ✗                       | ✗                       | ✓                 |
| TTS for pronunciation                               | ✓                     | ✗                            | ✓                       | ✗                       | ✓                 |
| Child-focused bilingual learning (under 8 years)    | ✓                     | ✗                            | ✓                       | ✗                       | ✓                 |
| Integration of NMT / AI translation                 | ✗                     | ✗                            | ✗                       | ✓                       | ✓                 |
| Evaluation with children (usability and engagement) | ✓                     | ✗                            | ✓                       | ✗                       | ✓                 |
| Gamified reward system (points, badges, word hunts) | ✓                     | ✓                            | ✗                       | ✗                       | ✓                 |

Table 1: Research Gap Comparison

### 1.3 Research Problem

Sinhala-speaking primary school children often struggle to understand and memorize Tamil words due to the lack of bilingual, child-centered, and interactive learning tools. Current resources either focus only on Tamil literacy or rely on static translation without engagement, as highlighted by Safeek and Ponniah [3] and further supported by Chowdhury and colleagues [8].

This project addresses that problem by developing a Bilingual Word Translation Module within the Hello Tamil system. The module will enable children to translate Tamil words into Sinhala and English using text and image input, supported by OCR for printed words, TTS for pronunciation, and gamified elements such as word hunts, points, and badges.

By integrating AI translation technologies-including NMT, OCR, and TTS with game-based learning principles, the proposed system aims to make Tamil language learning engaging, bilingual, and effective for non-native children in Sri Lanka, as demonstrated in the work of Choudhary and colleagues [4], Mei and co-authors [13], and Husin and Ismail [7].

## 2. Objectives

This section defines the primary aim and specific objectives of the proposed Bilingual Word Translation Module under the Hello Tamil - Tamil Language Learning System for Non-Native Children in Sri Lanka.

The objectives were formulated based on the identified research gap and are designed to guide the development, testing, and evaluation of the module.

### 2.1 Main Objective

To develop an interactive, gamified bilingual word translation module that enables Sinhala-speaking children under 8 years old to learn Tamil vocabulary by translating Tamil words into Sinhala and English, supported by text and image inputs, TTS pronunciation, and reward-based gamification mechanisms.

### 2.2 Specific Objectives

The specific objectives of this research are to:

1. To design and implement a bilingual Tamil–Sinhala/English translation system using Google Translate API or NMT for accurate word-level translation.
2. To integrate OCR technology to recognize and extract printed or handwritten Tamil words from images for translation.
3. To incorporate TTS functionality to help learners hear the correct pronunciation of translated words.
4. To develop gamified learning activities (e.g., Word Hunts, 2 Pictures 1 Meaning, Badge Collections) to motivate and engage children.
5. To evaluate system usability and learning effectiveness through pilot testing with non-native children and collect feedback from parents and teachers.
6. To ensure the system supports accessibility by designing a simple, child-friendly UI with visual cues and minimal text dependency.

| Objective                                                                                                    | Expected Outcome                                                                                                     |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Design and implement a bilingual Tamil–Sinhala/English translation system using Google Translate API or NMT. | Functional translation feature capable of accurately converting Tamil words into Sinhala and English for children.   |
| Integrate OCR technology to extract printed or handwritten Tamil words from images.                          | Working image-to-text conversion that allows users to learn vocabulary through photos or visual materials.           |
| Incorporate TTS functionality for pronunciation support.                                                     | Clear and child-friendly audio pronunciation of Tamil, Sinhala, and English words, enhancing phonetic understanding. |
| Develop gamified learning activities such as Word Hunts, 2 Pictures 1 Meaning, and Badge Collections.        | Engaging gameplay motivates children to practice and memorize new words through rewards and progress tracking.       |
| Evaluate system usability and learning effectiveness with non-native children.                               | Empirical evidence of improved vocabulary retention, engagement, and usability from pilot testing and feedback.      |
| Design a child-friendly, accessible interface with visual cues and minimal text dependency.                  | Simple, intuitive UI suitable for children under 8, improving interaction and learning comfort.                      |

*Table 2 Objectives and Expected Outcomes*

### 3. Methodology

#### 3.1 Key Pillars & Application of Technologies

The proposed system is built upon five key technological and pedagogical pillars that collectively support bilingual learning and gamified education for children:

1. OCR - Converts printed or handwritten Tamil text from images into editable text format for translation.
2. NMT - Translates recognized Sinhala or English into Tamil text using the Google Translate API or an open-source bilingual model.
3. TTS - By providing a clear audio reading of Tamil, Sinhala, and English words, TTS enhances vocabulary acquisition and improves comprehension, helping children to understand the meaning of new and unfamiliar terms as they hear them spoken.
4. Gamification Framework - Incorporates learning-based games, points, badges, rewards, word-hunt activities, and interactive challenges to increase motivation and engagement.
5. Child-Centered UX Design - Implements simple navigation, colorful visuals, image-based word association, and age-appropriate interactions to support young learners and improve vocabulary retention.

#### 3.2 Proposed System Architecture & Workflow

The system follows a structured workflow that integrates OCR, translation, pronunciation, and gamified learning into a single bilingual learning module:

##### Step 1: Input Acquisition

Children can enter Tamil words through:

- **Text input** (typing),
- **Image-based input** (captured or uploaded images), or
- **Preloaded word lists** within the app.



## **Step 2: OCR Processing**

Uploaded images undergo OCR to extract Tamil text, making printed or handwritten words usable for further processing.

## **Step 3: Translation Pipeline (NMT)**

Extracted or typed Tamil words are processed through a bilingual translation engine to generate:

- Sinhala translations
- English translations
- Optional transliteration for pronunciation clarity

## **Step 4: TTS Pronunciation Output**

For all three languages (Tamil, Sinhala, English), the TTS module offers:

- Word-level pronunciation
- Slow-mode reading for children
- Repeat-audio interaction

## **Step 5: Gamified Learning Layer**

Using the translated vocabulary, the system generates interactive learning activities such as:

- Word-hunt games
- Matching exercises
- Daily word challenges
- Points, badges, and progress tracking

## **Step 6: Feedback & Reinforcement**

Children receive immediate feedback (audio, animations, scores) to reinforce correct answers and improve long-term vocabulary retention.

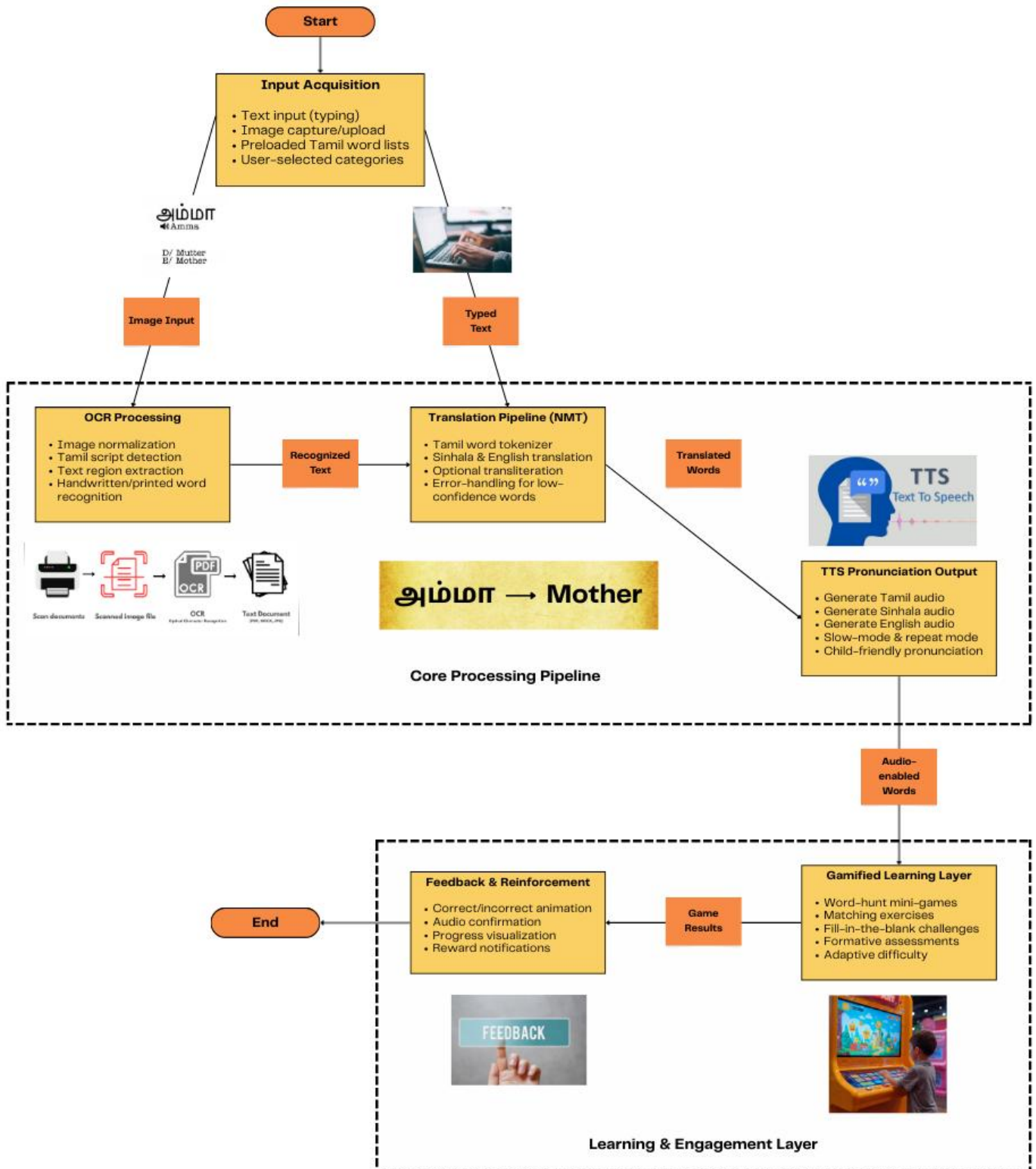


Figure 1: Proposed System Architecture

### 3.3 Data Sources & Preprocessing

The system primarily relies on Tamil word datasets, bilingual dictionaries, and image-based Tamil text samples collected from early-grade school materials and open educational repositories.

These datasets will serve as the foundation for training, testing, and validating the OCR and translation functionalities.

Data Sources:

- Tamil vocabulary datasets from Grade 1–3 government syllabi.
- Sinhala and English translations collected using verified bilingual dictionaries and Google Translate API.
- Image dataset containing printed and handwritten Tamil words (captured from textbooks, flashcards, and worksheets).

Preprocessing Steps:

1. Text normalization: Removing punctuation, special symbols, and redundant spaces from Tamil text.
2. Font unification: Converting mixed Tamil fonts into Unicode-compatible text for accurate OCR recognition.
3. Dataset labeling: Pairing each Tamil word with its Sinhala and English equivalents.
4. Image enhancement: Applying grayscale conversion, noise removal, and thresholding to improve OCR detection accuracy.
5. Data validation: Ensuring all Tamil words correspond correctly with their translated outputs.

The processed dataset will ensure that both the OCR and translation modules operate efficiently with minimal recognition errors and maximum language consistency.

### 3.4 OCR, Translation & TTS Integration

The translation pipeline consists of three main interconnected subsystems — OCR, Translation, and TTS — that function together to deliver bilingual learning support.

#### 1. OCR:

- Utilizes Tesseract OCR or Google Vision API to detect and extract Tamil text from image inputs.
- Converts visual Tamil words into Unicode text for further processing.
- Accuracy is enhanced through preprocessing and font normalization.

#### 2. Translation Engine:

- Employs Google Translate API or a fine-tuned NMT model to convert Tamil text into Sinhala and English.
- Ensures word-level translation optimized for early learners.
- Incorporates simplified vocabulary filtering to ensure age-appropriate translations.

#### 3. TTS:

- Uses Google TTS API or gTTS library to generate pronunciation audio for Tamil, Sinhala, and English outputs.
- Audio playback is embedded in the interface, allowing children to listen and repeat.

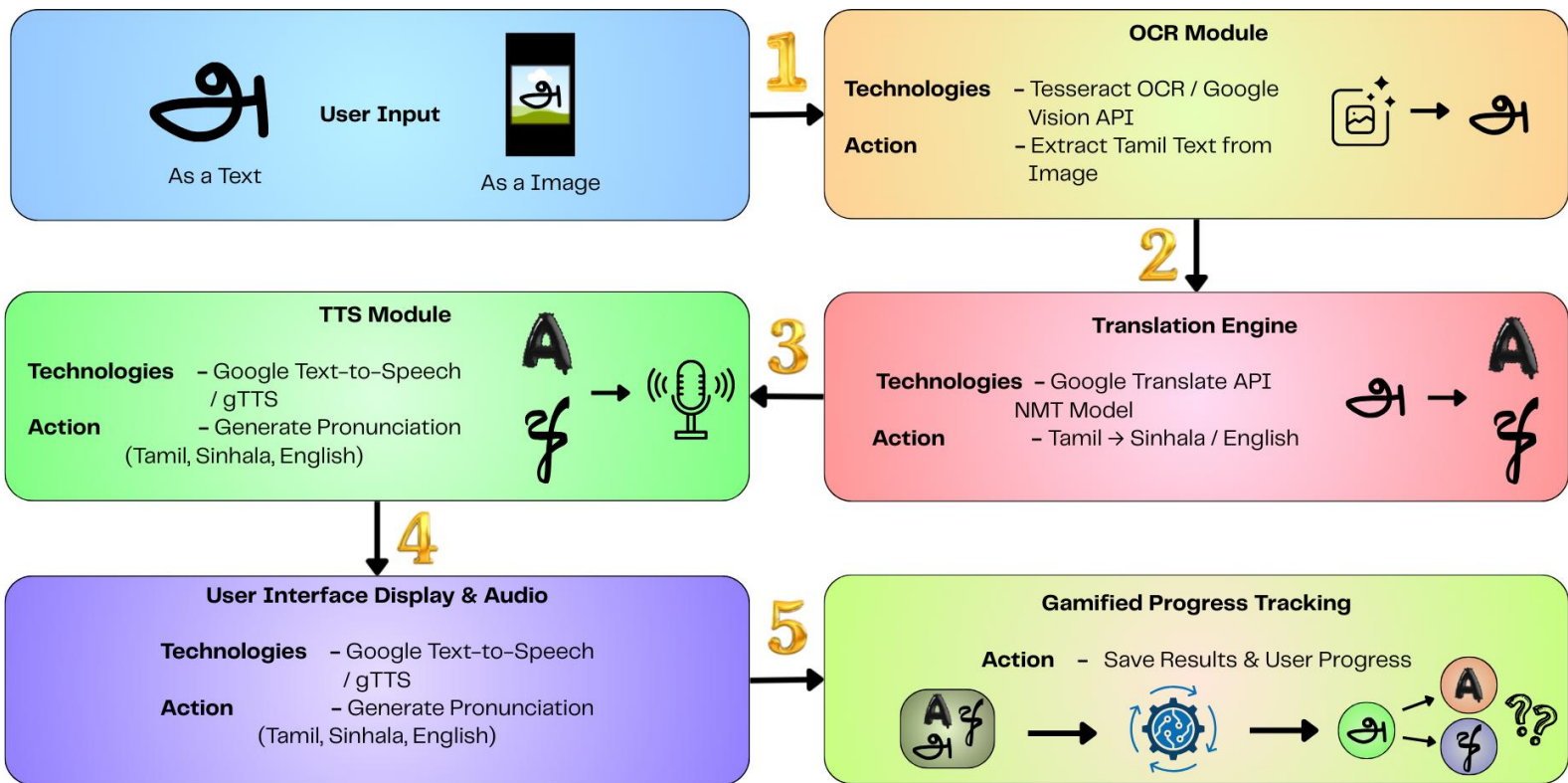


Figure 2: OCR and Translation Workflow

### 3.5 Gamification Design & Reward Mechanisms

To make language learning enjoyable and effective for young users, the system integrates gamification principles into its learning activities.

Gamification transforms translation practice into a series of fun, reward-based challenges that sustain motivation and learning interest.

#### Key Game Elements:

- **Word Hunt Game:** Children complete daily word-hunt challenges by translating a given number of Tamil words correctly within 24 hours.
- **2 Pictures 1 Meaning Game:** Inspired by Husin & Ismail (2025), children match two visual clues to the correct translation, reinforcing bilingual understanding.
- **Badge System:** Learners earn badges and points upon reaching milestones (e.g., 10, 20, 50 words learned).
- **Progress Bar:** Displays vocabulary mastery progress to encourage continued participation.
- **Reward Points (XP):** Accumulated for successful translations; can be used to unlock new levels or word sets.

#### Educational Benefits:

- Promotes intrinsic motivation through exploration and achievement.
- Reinforces repetition-based learning for vocabulary retention.
- Enhances self-paced learning without external pressure.

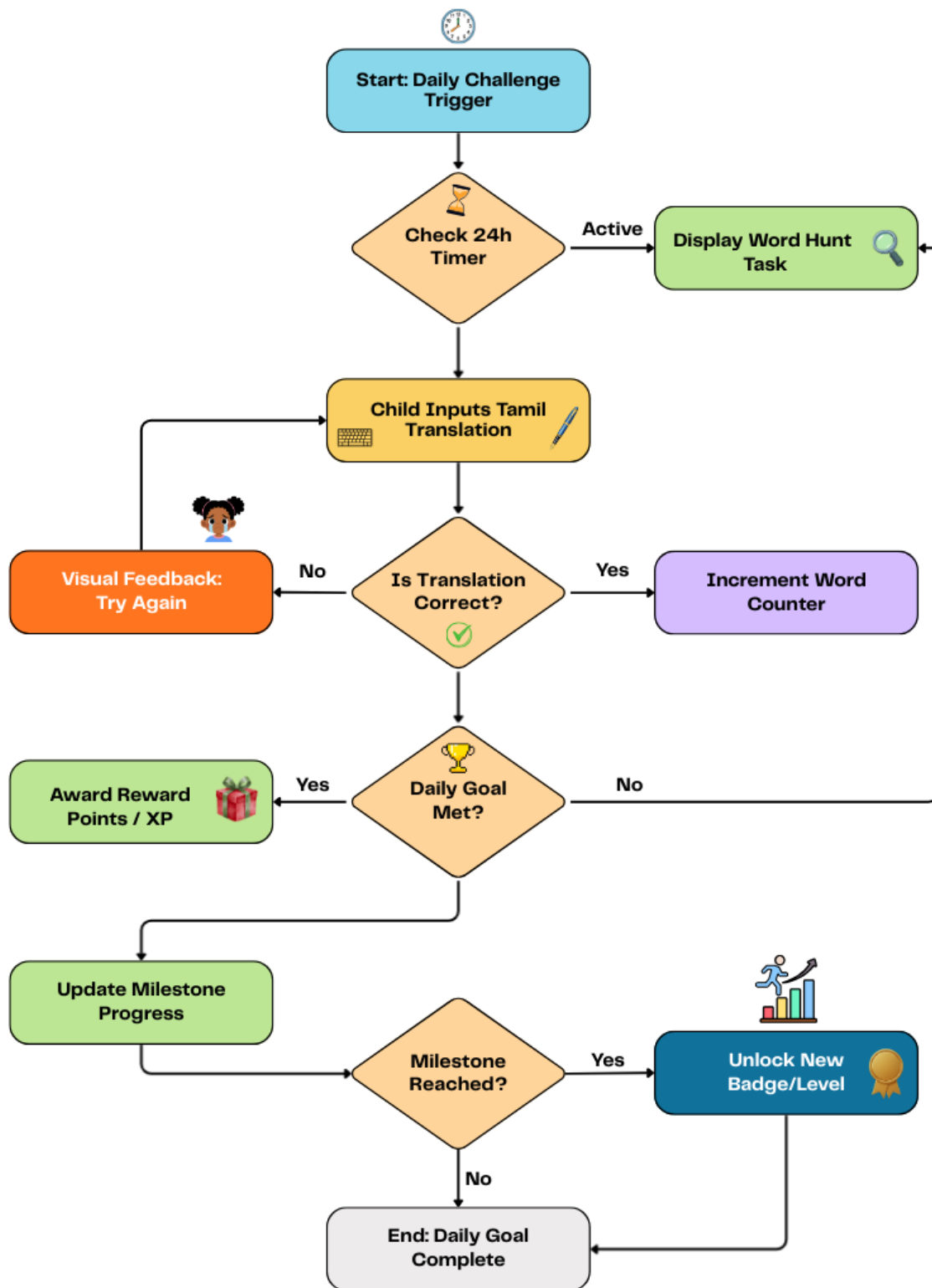


Figure 3: Gamified Word Hunt Workflow

### 3.6 Tools & Technologies

| Category             | Technology / Tool                 | Purpose / Description                                           |
|----------------------|-----------------------------------|-----------------------------------------------------------------|
| Programming Language | Python, JavaScript                | For backend (translation & OCR logic) and frontend (UI design). |
| Mobile Framework     | Flutter / React Native            | Cross-platform mobile app development (Android & iOS).          |
| Database             | Firebase / MongoDB                | Store user profiles, progress, and translated words.            |
| OCR Tool             | Tesseract OCR / Google Vision API | Extract Tamil text from images.                                 |
| Translation Engine   | Google Translate API / OpenNMT    | Tamil → Sinhala / English translation.                          |
| TTS Engine           | Google TTS (gTTS)                 | Generate Tamil, Sinhala, and English pronunciations.            |
| Cloud Services       | Google Cloud Platform             | Host APIs and manage OCR, TTS, and translation services.        |
| Gamification Logic   | Custom XP system                  | Track user progress, points, and rewards.                       |
| Version Control      | GitHub                            | Code management and version tracking.                           |
| UI/UX Design         | Figma / Canva                     | Design child-friendly, colorful, and interactive interfaces.    |

*Table 3: Tools & Technologies*



### 3.7 Project Requirements

#### 3.7.1 Functional Requirements (FR)

Functional requirements define the specific operations and behaviors that the system must perform to meet user and project objectives.

The following table lists the key functional requirements of the module.

1. User Registration & Profile Management - The system shall allow children to register with basic details or use a parent-supervised account to track learning progress.
2. Text Input Translation - The system shall allow users to input Tamil text and receive accurate Sinhala and English translations.
3. Image Input via OCR - The system shall allow users to upload or capture images containing Tamil text and convert it to digital text using OCR for translation.
4. TTS Pronunciation - The system shall generate and play audio pronunciation for Tamil, Sinhala, and English words.
5. Gamified Learning Activities - The system shall provide mini-games such as *Word Hunts* and *2 Pictures 1 Meaning* to enhance learning engagement.
6. Progress Tracking & Rewards - The system shall track completed lessons, total words learned, and award points, badges, or stars based on user performance.
7. Word Collection Library - The system shall maintain a personal vocabulary list of translated words for each user to review later.
8. Daily Learning Challenge - The system shall generate a daily or weekly word hunt challenge to motivate consistent learning.
9. Parent/Teacher Feedback Module - The system shall allow parents or teachers to view child progress reports and learning statistics.
10. Data Storage and Synchronization - The system shall store all translation data, game scores, and progress in a secure database with real-time synchronization.

### 3.7.2 Non-Functional Requirements

Non-functional requirements define the performance, reliability, and usability standards that ensure the system's effectiveness and user satisfaction.

1. Usability - The interface shall be simple, colorful, and child-friendly, with icons and visuals suitable for users under 8 years old.
2. Performance - The system shall process translation requests and OCR recognition within 3 seconds for a smooth UX
3. Reliability - The application shall ensure consistent translation accuracy and maintain stable operation under varying network conditions.
4. Scalability - The system shall be easily extendable to support additional languages or new gamified features.
5. Security & Privacy - User data (profiles, progress, and recordings) shall be securely stored and accessible only to authorized users.
6. Compatibility - The application shall operate on Android and iOS platforms with responsive layouts for various screen sizes.
7. Maintainability - The system architecture shall be modular, enabling updates or improvements without affecting other components.
8. Accessibility - The system shall include visual and auditory cues to support children with reading or learning difficulties.
9. Localization - The system shall support multilingual interfaces (Tamil, Sinhala, English) for ease of understanding.
10. Availability - The application shall maintain at least 95% uptime for online translation and gameplay services.

### 3.8 Work Breakdown Structure (WBS)

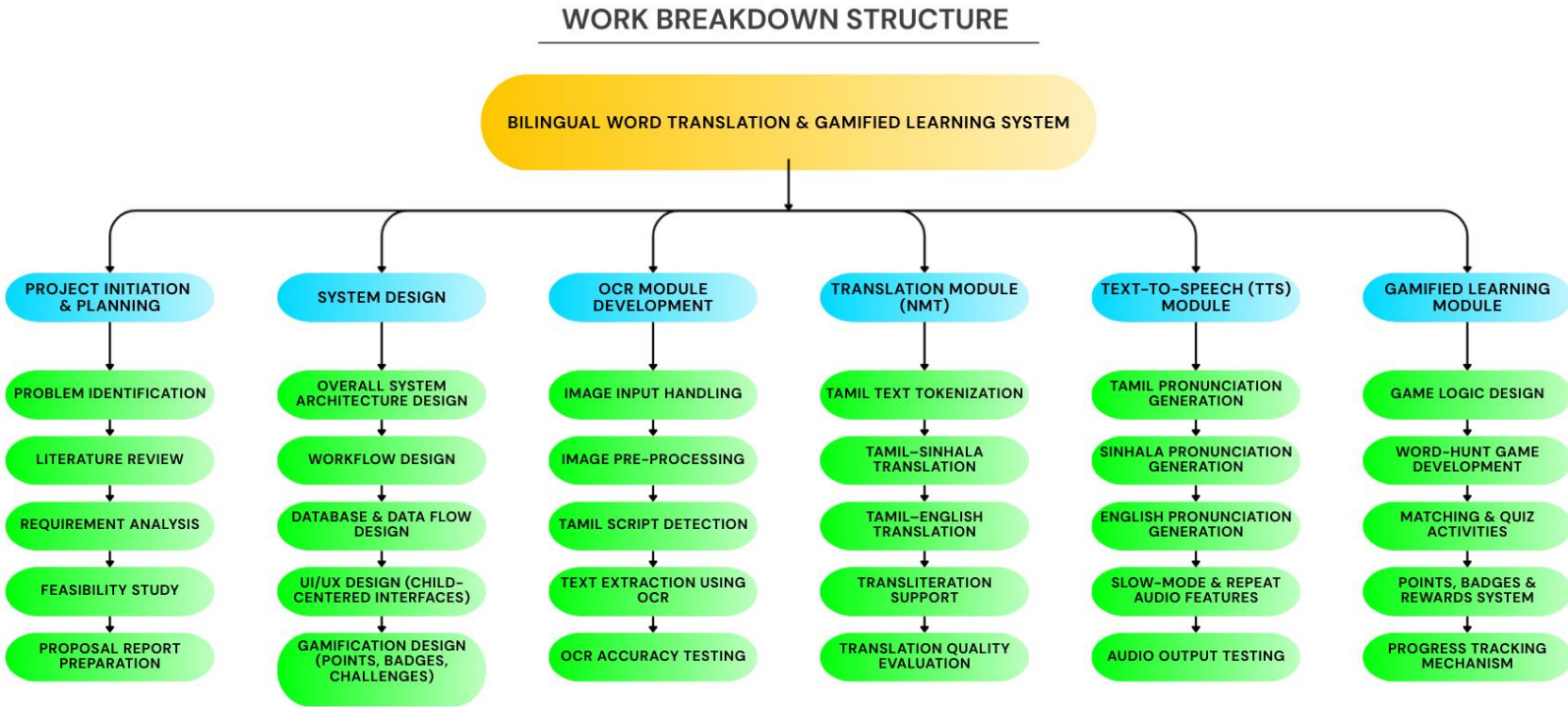


Figure 3: Work Breakdown Structure

### 3.9 Gantt Chart

A Gantt chart will be created to visualize the project timeline, mapping activities against weeks. Overlapping phases (e.g., documentation and testing) will ensure efficient project progress.

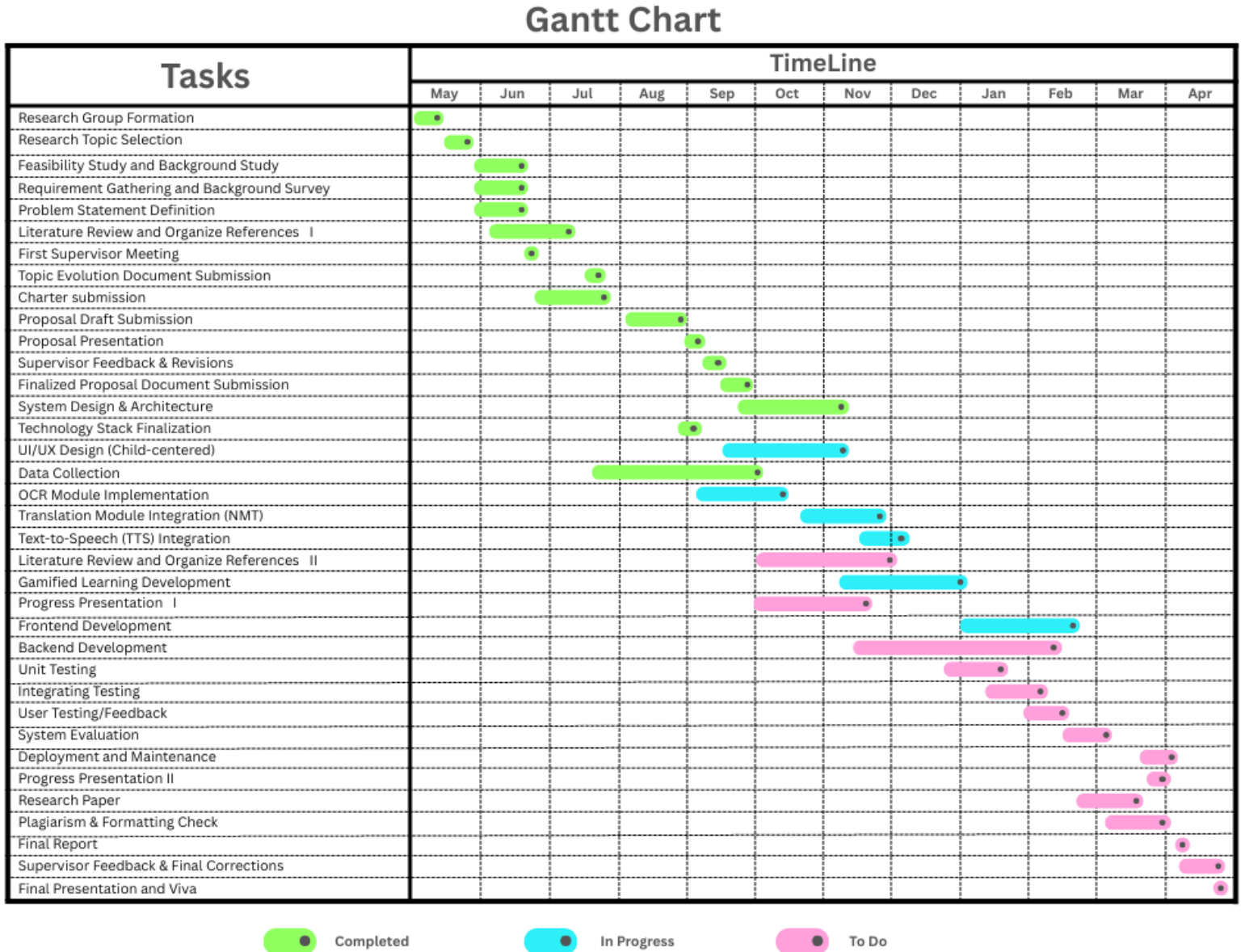


Figure 5: Gantt Chart

## 4. DESCRIPTION OF PERSONNEL AND FACILITIES

### 4.1 Personnel

The successful completion of this module relies on the efforts of the student researcher, supervisor, and co-supervisor. Their respective responsibilities are outlined in **Table 3** below.

| Name / Role             | Designation        | Responsibilities                                                                                                                                                                                         |
|-------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ahamed M.A.S.           | Student Researcher | Responsible for requirement analysis, system design, development of the translation and gamification features, integration of OCR and TTS modules, testing, documentation, and final report preparation. |
| Dr. Mr. Suresh Niroshan | Supervisor         | Provides technical guidance, supervises project progress, assists with system design validation, and ensures academic and research standards are maintained.                                             |
| Ms. Fathima Fanoon      | Co-Supervisor      | Supports evaluation of the translation accuracy, usability, and linguistic quality of the Tamil–Sinhala outputs. Offers feedback on UI/UX design and educational impact.                                 |

*Table 4: Roles and Responsibilities*

Additionally, external reviewers and participants (teachers or parents) will contribute during the evaluation phase, helping test the usability and educational effectiveness of the system with non-native children.

## 4.2 Facilities

The development and testing of the proposed system will be conducted using the computing resources, software environments, and laboratory facilities available through the institution. The facilities required for successful completion are summarized in Table 4 below.

| Facility Type              | Description / Purpose                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Computer Laboratory Access | Used for software development, debugging, and testing of the OCR, translation, and gamification components.           |
| Internet Connectivity      | Required for API integration (Google Translate, OCR, and TTS) and online database synchronization.                    |
| Software Tools             | Python, React Native, Firebase, Google Vision API, Google Translate API, Google TTS, and Figma for UI/UX design.      |
| Mobile Testing Devices     | Android smartphones and tablets for testing the responsiveness and usability of the mobile application.               |
| Cloud Storage and Hosting  | Google Cloud Platform for hosting APIs, database storage, and real-time user analytics.                               |
| Reference Materials        | Access to research databases, e-journals, and online resources for continued literature review and technical updates. |

*Table 5: Project Resources and Facilities*

All required facilities are readily available within the institutional computing environment or accessible via cloud-based tools, ensuring a smooth development and testing workflow throughout the project lifecycle.

## 5. Budget and Budget Justification

### 5.1 Budget Estimate

| No.                        | Item Description                                                 | Category      | Estimated Cost (LKR) | Remarks                                             |
|----------------------------|------------------------------------------------------------------|---------------|----------------------|-----------------------------------------------------|
| 1                          | Computer / Laptop (personal system for development)              | Hardware      | 0.00                 | Already available personal device                   |
| 2                          | Internet Connectivity (API access, online testing, hosting)      | Utility       | 3,000.00             | Monthly data usage for testing and deployment       |
| 3                          | Google Cloud Services (API usage for Translate, Vision, and TTS) | Cloud Service | 5,000.00             | Pay-as-you-go API usage for OCR and translation     |
| 4                          | Firebase Database & Hosting (Cloud storage and analytics)        | Cloud Service | 2,000.00             | Cloud-based backend hosting and synchronization     |
| 5                          | Figma / Canva (UI/UX design tools)                               | Software Tool | 1,500.00             | Used for designing child-friendly app interfaces    |
| 6                          | Mobile Devices for Testing (Android smartphone/tablet)           | Hardware      | 0.00                 | Borrowed from institutional facility                |
| 7                          | Miscellaneous (Printing, documentation, backup storage)          | Other         | 1,500.00             | For printing reports, charts, and evaluation sheets |
| 8                          | Pilot Testing Materials (word cards, feedback forms)             | Evaluation    | 1,000.00             | Required for child-based learning evaluation        |
| 9                          | Travel / Communication (data collection, school visits)          | Logistics     | 2,000.00             | For data gathering and testing coordination         |
| 10                         | Contingency                                                      | Miscellaneous | 1,000.00             | Buffer for unexpected project-related expenses      |
| Total Estimated Cost (LKR) |                                                                  |               | 17,000.00            |                                                     |

*Table 6: Budget Distribution*

## **5.2 Budget Justification**

- The proposed budget is minimal since most hardware and software resources are already available or offered by the institution.
  - Internet and cloud services are required for API integration (Google Translate, Vision, and TTS) and data synchronization.
  - A small allocation is made for Firebase hosting, Figma/Canva design tools, and printing materials used in testing.
  - Travel and communication costs cover limited school visits for evaluation with children.
  - A contingency amount is included for unexpected expenses.
- Overall, the total estimated cost of LKR 17,000.00 is sufficient to complete the project effectively and meet all objectives.



## 6. Anticipated Conclusion

### 6.1 Anticipated Results

The proposed module is expected to produce the following key outcomes:

1. **Functional Mobile Application:**  
A fully operational bilingual Tamil–Sinhala/English translation app that allows text and image-based translation, pronunciation, and interactive learning.
2. **Enhanced Vocabulary Learning:**  
Children will demonstrate improved understanding and retention of Tamil vocabulary through translation, image association, and gamified activities.
3. **Effective Use of Technology:**  
Successful integration of OCR, Translation APIs, and TTS will showcase how AI can be applied in early language learning contexts.
4. **Child Engagement through Gamification:**  
The inclusion of reward-based games (e.g., Word Hunts, 2 Pictures 1 Meaning) will foster higher motivation, enjoyment, and consistent participation in learning.
5. **Evaluation Outcomes:**  
Feedback collected from children, parents, and teachers is expected to confirm improvements in usability, engagement, and vocabulary recall.
6. **Educational Impact:**  
The results are anticipated to contribute to bilingual education by providing an accessible tool for Tamil learning among non-native Sinhala-speaking children in Sri Lanka.

## 6.2 Conclusion

The Bilingual Word Translation Module serves as a crucial step toward bridging the linguistic gap between Tamil and Sinhala learners at the primary education level. By integrating AI-based translation, OCR, TTS, and gamification, this project transforms traditional rote learning into a fun, interactive, and bilingual digital experience.

The expected outcome is an innovative learning solution that not only improves Tamil vocabulary acquisition but also promotes cross-linguistic understanding and cultural inclusivity among young learners. Furthermore, the research findings can serve as a foundation for future expansions of the Hello Tamil system—such as sentence-level translation, advanced pronunciation training, and AI-driven personalized learning pathways.

In conclusion, this project aims to contribute both academically and socially by developing a meaningful technological tool that makes Tamil learning engaging, bilingual, and accessible for every Sri Lankan child.

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## 8. Appendices

### 8.1 Appendix A: Sample Tamil–Sinhala & English Word Dataset

The following table presents a small sample of the **Tamil–Sinhala–English word dataset** used in the system for translation and learning purposes. Each record contains a Tamil word, its Sinhala and English meanings, and associated category (for lesson grouping in the app).

| Tamil Word          | Sinhala Translation | English Translation | Category   |
|---------------------|---------------------|---------------------|------------|
| அப்பா (Appa)        | නාන්තා              | Father              | Family     |
| அம்மா (Amma)        | අම්මා               | Mother              | Family     |
| நீர் (Neer)         | වතුර                | Water               | Daily Life |
| பூ (Poo)            | මල                  | Flower              | Nature     |
| வீடு (Veedu)        | ගෙදර                | House               | Places     |
| புத்தகம் (Puthagam) | පොත                 | Book                | Education  |
| மழை (Mazhai)        | වැසි                | Rain                | Weather    |
| நாய் (Naai)         | බේලා                | Dog                 | Animals    |
| பல் (Pal)           | දන්                 | Tooth               | Body       |
| சூரியன் (Sooriyan)  | හිරු                | Sun                 | Nature     |

### 8.2 Appendix B: Sample Word Translation Dataset

#### Example 1:

Tamil Word: அப்பா (Appa)

Sinhala Translation: නාන්තා

English Translation: Father

Context: Common family-related vocabulary used in early-grade Tamil lessons.

#### Example 2:

Tamil Word: மழை (Mazhai)

Sinhala Translation: වැසි

English Translation: Rain

Context: Basic weather vocabulary used in daily communication activities.

**Example 3:**

Tamil Word: புத்தகம் (Puthagam)

Sinhala Translation: පොත

English Translation: Book

Context: Educational vocabulary used in school-related scenarios.

**Example 4:**

Tamil Word: நாய் (Naai)

Sinhala Translation: බේලා

English Translation: Dog

Context: Common animal name used in beginner reading materials.

**Example 5:**

Tamil Word: மலர் (Malar)

Sinhala Translation: මල

English Translation: Flower

Context: Nature-based vocabulary included in story and image games.

**Example 6:**

Tamil Word: வீடு (Veedu)

Sinhala Translation: ගෙදර

English Translation: House

Context: Common location-related term used in visual matching and word hunt games.

### 8.3 Turnitin Report